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United States Patent	12411092
Kind Code	B2
Date of Patent	September 09, 2025
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Optical monitoring device

Abstract

The present disclosure provides optical monitoring devices. In some embodiments, the optical monitoring device includes a heating stirrer, a floating barrel, and a support body arranged to support the floating barrel. The floating barrel includes a barrel body facing the heating stirrer and spaced apart from the heating stirrer by a gap, a cavity provided in the barrel body and configured to accommodate at least one of a vessel and a conduit, an irradiator arranged in the barrel body and configured to radiate light to the cavity, and a light receiver arranged in the barrel body, aligned with the irradiator and the cavity, and configured to receive light from the cavity.

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Appl. No.: 18/140406

Filed: April 27, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240201093 A1	Jun. 20, 2024

Foreign Application Priority Data

KR	10-2022-0175583	Dec. 15, 2022
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Publication Classification

Int. Cl.: G01N21/85 (20060101); G01N21/3577 (20140101)

U.S. Cl.:

CPC G01N21/85 (20130101); G01N21/3577 (20130101); G01N2021/8557 (20130101); G01N2201/0231 (20130101); G01N2201/061 (20130101)

Field of Classification Search

CPC: G01N (21/85); G01N (21/3577); G01N (2021/8557); G01N (2201/0231); G01N (2201/061); G01N (21/75); G01N (2021/8416); G01N (2201/08); G01N (2201/1296); G01N (21/272); G01N (21/31); G01N (21/25); B01F (33/452); G02B (7/02); G02B (26/001)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims benefit of priority under 35 USC § 119 of Korean Patent Application No. 10-2022-0175583, filed on Dec. 15, 2022, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

(2) The present disclosure relates generally to an optical monitoring device.

2. Description of Related Art

(3) A device has been developed for analyzing a pattern of light that passes through, or is deflected or scattered from a liquid sample. The device may detect the light that is not absorbed by the liquid sample. When a path of light between a light source of the device, the liquid sample, and a detector of the device is precisely set, the device may produce repeatable and iterative measurement results.

SUMMARY

(4) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

(5) According to an aspect of the present disclosure, an optical monitoring device is provided. The optical monitoring device includes a heating stirrer, a floating barrel, and a support body arranged to support the floating barrel. The floating barrel includes a barrel body facing the heating stirrer and spaced apart from the heating stirrer by a gap, a cavity provided in the barrel body and configured to accommodate at least one of a vessel and a conduit, an irradiator arranged in the barrel body and configured to radiate light to the cavity, and a light receiver arranged in the barrel body, aligned with the irradiator and the cavity, and configured to receive light from the cavity.

(6) In some embodiments, in the optical monitoring device a size of the gap between the barrel body and the heating stirrer may be greater than or equal to approximately 3 mm.

(7) In some embodiments, the support body of the optical monitoring device may include a holder configured to hold the floating barrel, a plurality of guide rails configured to guide movement of the holder, and a plurality of legs coupling the holder to the plurality of guide rails.

(8) In some embodiments, the support body of the optical monitoring device may further include a knob configured to adjust a position of a leg of the plurality of legs on a corresponding guide rail of the plurality of guide rails.

(9) In some embodiments, each of the plurality of legs may include a tab hole configured to pass a corresponding guide rail of the plurality of guide rails.

(10) In some embodiments, the heating stirrer may include a heating plate facing the barrel body, and a jig protruding from the heating plate, and configured to separate the at least one of a vessel and a conduit, from the barrel body.

(11) In some embodiments, the jig may include a first slit aligned with the irradiator, and a second slit aligned with the light receiver.

(12) In some embodiments, the barrel body may include an insulating material.

(13) In some embodiments, the barrel body may include a first passage extending from the cavity and configured to accommodate the irradiator, and a second passage extending from the cavity and configured to accommodate the light receiver.

(14) In some embodiments, each of the irradiator and the light receiver may include at least one

lens, and a lens housing configured to accommodate the at least one lens. The lens housing may be further configured to be detachably coupled to the barrel body.

(15) In some embodiments, the at least one lens may include a converging lens and a diverging lens.

(16) In some embodiments, the optical monitoring device may further include a light source configured to generate light in a determined wavelength band, and a spectroscope configured to analyze at least a portion of the light generated by the light source and received by the light receiver that has been radiated by the irradiator and has passed through the at least one of a vessel and a conduit.

(17) In some embodiments, the optical monitoring device may further include a chiller configured to cool the at least one of a vessel and a conduit.

(18) In some embodiments, the optical monitoring device may further include a circulator configured to circulate a cooling liquid to the chiller.

(19) In some embodiments, the barrel body may further include an access area open to the cavity.

(20) In some embodiments, the optical monitoring device may further include a shaft, and an arm configured to connect the shaft to the floating barrel and rotate relative to the shaft.

(21) In some embodiments, the optical monitoring device may further include a first linear guide configured to guide linear movement of a first lens of the irradiator in a first direction toward the cavity and a second direction away from the cavity, and a second linear guide configured to guide linear movement of a second lens the light receiver in the first direction toward the cavity and the second direction away from the cavity.

(22) In some embodiments, the optical monitoring device may further include a tab configured to secure a lens of at least one of the irradiator and the light receiver, and an elastic body configured to elastically support the tab.

(23) In some embodiments, the optical monitoring device may further include a slider configured to slide in a first direction away from the lens, slide in a second direction toward the lens, and deform the elastic body.

(24) In some embodiments, the optical monitoring device may further include a handle coupled to the slider.

(25) Additional aspects are set forth in part in the description which follows and, in part, may be apparent from the description, or may be learned by practice of the presented embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects, features, and advantages of certain embodiments of the present disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a block diagram illustrating an optical monitoring device, according to an embodiment;

(3) FIG. 2 is a perspective view of an optical monitoring device, according to an embodiment;

(4) FIG. 3 is a perspective view of a floating barrel, according to an embodiment;

(5) FIG. 4 is a plan view of the floating barrel, according to an embodiment;

(6) FIG. 5 is a side view of the floating barrel, according to an embodiment;

(7) FIG. 6 is a cross-sectional view of the floating barrel of FIG. 4 taken along a line 6-6, according to an embodiment;

(8) FIG. 7 is a cross-sectional view of the floating barrel of FIG. 5 taken along a line 7-7, according to an embodiment;

(9) FIG. 8 is a perspective view of a jig, according to an embodiment;

- (10) FIG. 9 is a diagram illustrating a floating barrel at a predetermined distance from a heating plate, according to an embodiment;
- (11) FIG. 10 is a diagram illustrating thermal deformation of a floating barrel at a predetermined distance from a heating plate, according to an embodiment;
- (12) FIG. 11 is a diagram illustrating a height deviation of lenses, according to an embodiment;
- (13) FIG. 12 are graphs showing a height deviation and a light loss factor based on a distance between a heating plate and a floating barrel, according to an embodiment;
- (14) FIG. 13 is a perspective view of a floating barrel, according to an embodiment;
- (15) FIG. 14 is a perspective view of a linear guide, according to an embodiment;
- (16) FIG. 15 is a perspective view of a structure for fixing and releasing a lens, according to an embodiment;
- (17) FIG. 16 is a perspective view of a structure for fixing and releasing a lens, according to an embodiment;
- (18) FIG. 17 is a diagram schematically illustrating an optical monitoring device, according to an embodiment; and
- (19) FIG. 18 is a graph showing a light loss factor based on a distance between a heating plate and a floating barrel, according to an embodiment.

DETAILED DESCRIPTION

(20) Hereinafter, example embodiments are described in detail with reference to the accompanying drawings. However, various alterations and modifications may be made to the example embodiments. That is, the example embodiments are not construed as limited to the disclosure. The example embodiments should be understood to include all changes, equivalents, and replacements within the idea and the technical scope of the disclosure.

(21) The terminology used herein is for the purpose of describing particular example embodiments only and is not to be limiting of the example embodiments. The singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It is to be understood that the terms “comprises/comprising” and/or “includes/including” when used herein, may specify the presence of stated features, integers, steps, operations, elements, and/or components, but may not preclude the presence and/or addition of one or more other features, integers, steps, operations, elements, components and/or groups thereof.

(22) Unless otherwise defined, all terms including technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which example embodiments belong. It is to be further understood that terms, such as those defined in commonly-used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and may not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(23) When describing the example embodiments with reference to the accompanying drawings, like reference numerals refer to like constituent elements and a repeated description related thereto may be omitted. The accompanying drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and/or convenience. In the description of example embodiments, detailed description of well-known related structures or functions may be omitted when it is deemed that such description may cause ambiguous interpretation of the present disclosure.

(24) Also, in the description of the components, terms such as first, second, A, B, (a), (b) or the like may be used herein when describing components of the present disclosure. These terms are used only for the purpose of discriminating one constituent element from another constituent element, and the nature, the sequences, or the orders of the constituent elements are not limited by the terms. When one constituent element is described as being “connected”, “coupled”, or “attached” to another constituent element, it should be understood that one constituent element can be connected or attached directly to another constituent element, and an intervening constituent element can also

be “connected”, “coupled”, or “attached” to the constituent elements.

(25) It is to be understood that when an element or layer is referred to as being “over,” “above,” “on,” “below,” “under,” “beneath,” “connected to” or “coupled to” another element or layer, it can be directly over, above, on, below, under, beneath, connected or coupled to the other element or layer or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly over,” “directly above,” “directly on,” “directly below,” “directly under,” “directly beneath,” “directly connected to” or “directly coupled to” another element or layer, there may no intervening elements or layers present. The terms “upper,” “middle”, “lower”, etc. may be replaced with terms, such as “first,” “second,” “third” to be used to describe relative positions of elements.

(26) Reference throughout the present disclosure to “one embodiment,” “an embodiment,” “an example embodiment,” or similar language may indicate that a particular feature, structure, or characteristic described in connection with the indicated embodiment is included in at least one embodiment of the present solution. Thus, the phrases “in one embodiment”, “in an embodiment,” “in an example embodiment,” and similar language throughout this disclosure may, but do not necessarily, all refer to the same embodiment.

(27) Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list. For example, the expression, “at least one of a, b, and c,” should be understood as including only a, only b, only c, both a and b, both a and c, both b and c, all of a, b, and c, or any variations of the aforementioned examples.

(28) The same name may be used to describe an element included in the example embodiments described above and an element having a common function. Unless otherwise mentioned, the descriptions on the example embodiments may be applicable to the following example embodiments and thus, duplicated descriptions may be omitted for conciseness.

(29) FIG. 1 is a block diagram illustrating an optical monitoring device, according to an embodiment.

(30) Referring to FIG. 1, an optical monitoring device **10** may be configured to measure, observe, and/or detect an in-situ reactant in a chemical reaction process. Light generated by a light source **11** may travel to a reactor **12** through at least one optical fiber F1 connecting the light source **11** to the reactor **12**. The light source **11** may be configured to generate light in one or more predetermined wavelength bands. In an embodiment, the light source **11** may be configured to generate light in at least one of an ultraviolet band, a visible band, and an infrared band. Alternatively or additionally, the light source **11** may be configured to generate a plurality of light rays in corresponding bands. For example, a first light in a first wavelength band generated by the light source **11** may travel to the reactor **12** through one optical fiber F1. Alternatively or additionally, a second light in a second wavelength band, that is different from the first wavelength band, and generated by the light source **11** may travel to the reactor **12** through another optical fiber F1. The light that traveled to the reactor **12** may penetrate (e.g., pass-through) and/or may be absorbed by a substance in the reactor **12**.

(31) The reactor **12** may be configured to provide thermal energy and/or stirring energy to an environment (e.g., a reaction vessel and/or a conduit) in which the substance is disposed. An in-situ chemical reaction process of a substance in the reactor **12** may be monitored in order to assess the stability of the chemical reaction and/or whether the chemical reaction is an intended reaction process. The monitoring and/or assessment of the chemical reaction process may include measuring and/or controlling of at least one a heat supply amount, a heat supply time, a temperature control time, other control parameters of the reactor **12**, and/or a combination thereof. For example, the control parameters may be determined to obtain a maximum efficiency and/or a maximum yield of the chemical reaction.

(32) In an embodiment, the light penetrating (e.g., passing through) the substance may travel to a spectroscope **13** through at least one optical fiber F2 connecting the reactor **12** to the spectroscope

13. The spectroscope **13** may be configured to obtain spectroscopic pattern data based on a substance by analyzing the penetrating light during the chemical reaction in the reactor **12** in real-time. In an optional or additional embodiment, the spectroscopic pattern data may be used as training data input to a neural network. Consequently, the spectroscopic pattern data and/or training result data thereof may be used to predict a chemical reaction mechanism.

(33) The optical monitoring device **10** may be applied to an application that monitors a chemical change of a substance in a vessel and/or a conduit for heating and/or stirring in real-time. For example, the optical monitoring device **10** may be applied to monitor a process of a pharmaceutical and/or chemical product that may need in-situ monitoring of a change in a component and/or a characteristic occurring in an environment in the unit of a vessel and/or a conduit. For another example, the optical monitoring device **10** may be applied to detect, identify, and/or select a target substance in a sealed state in a vessel and/or a conduit. For yet another example, the optical monitoring device **10** may be applied to a spectroscopic analysis using the light source **11** and the spectroscope **13**.

(34) In an embodiment, the optical monitoring device **10** may include a chiller **14**. The chiller **14** may be configured to chill a vessel and/or a conduit in the reactor **12** to maintain the vessel and/or the conduit in an intended temperature range. For example, the chiller **14** may maintain a temperature of the reactor **12** to be in an intended temperature range by exchanging heat with the reactor **12** through a cooling liquid. That is, the chiller **14** may be configured to prevent a level of a solution from decreasing as a solvent boils and/or evaporates in a vessel and/or a conduit while a chemical reaction is conducted in the reactor **12** for an extended period of time, for example.

(35) In an embodiment, the optical monitoring device **10** may include a circulator **15**. The circulator **15** may form a circulation path of a cooling liquid between the chiller **14** and the circulator **15**. The circulator **15** may provide a cooling liquid to the chiller **14** and/or may be provided with a cooling liquid by the chiller **14**.

(36) FIG. 2 is a perspective view of an optical monitoring device, according to an embodiment. FIG. 3 is a perspective view of a floating barrel, according to an embodiment. FIG. 4 is a plan view of the floating barrel, according to an embodiment. FIG. 5 is a side view of the floating barrel, according to an embodiment. FIG. 6 is a cross-sectional view of the floating barrel of FIG. 4 taken along a line 6-6, according to an embodiment. FIG. 7 is a cross-sectional view of the floating barrel of FIG. 5 taken along a line 7-7, according to an embodiment.

(37) Referring to FIGS. 2 to 7, an optical monitoring device **100** may include a reactor **101**. The reactor **101** may allow a chemical reaction of a substance in a reaction vessel **102** to proceed.

(38) In an embodiment, the reactor **101** may include a heating stirrer **110**. The heating stirrer **110** may be configured to generate thermal energy and/or stirring energy.

(39) In an embodiment, the heating stirrer **110** may include a driving unit **111**. The driving unit **111** may include a cylindrical shape. The driving unit **111** may include a magnetic bar for providing a stirring action to a liquid in the driving unit **111**. The driving unit **111** may further include a magnet and/or an electromagnetic element (e.g., a coil) configured to drive a motion of the magnetic bar. The driving unit **111** may further include a heating element configured to generate thermal energy.

(40) In an embodiment, the heating stirrer **110** may include a heating plate **112**. The heating plate **112** may have a substantially circular and/or elliptical cross-section. The heating plate **112** may be disposed on the driving unit **111**. The heating plate **112** may include, but not be limited to, a metal material (e.g., aluminum). The heating plate **112** may be physically connected to the driving unit **111**. Alternatively or additionally, the heating plate **112** may be thermally coupled to the driving unit **111**. That is, when vibration energy and thermal energy generated by the driving unit **111** are transferred to the heating plate **112**, the heating plate **112** may be configured to vibrate and transfer the thermal energy to the reaction vessel **102**.

(41) In an embodiment, the reactor **101** may include a floating barrel **120**. The floating barrel **120** may be configured to float from the heating stirrer **110**. The floating barrel **120** may be isolated

from the thermal energy and/or the vibration energy generated by the heating stirrer **110**.

(42) The floating barrel **120** may include a barrel body **121**. The barrel body **121** may include a first barrel surface **121A** (e.g., a top barrel surface), a second barrel surface **121B** (e.g., a bottom barrel surface) opposite to the first barrel surface **121A**, and a plurality of side barrel surfaces **121C** between the first barrel surface **121A** and the second barrel surface **121B**. The barrel body **121** may have a substantially hexahedral shape having four side barrel surfaces **121C**. Alternatively or additionally, the barrel body **121** may have a substantially cylindrical shape having one side barrel surface **121C**.

(43) The barrel body **121** may face the heating stirrer **110**. The second barrel surface **121B** may face a surface (e.g., a surface in a +Z normal direction) of the heating plate **112**. The barrel body **121** may be physically separated and/or spaced apart from the heating stirrer **110** with a gap G. The size of the gap G may refer to a distance between the second barrel surface **121B** and a surface (e.g., the surface in the +Z normal direction) of the heating plate **112** facing the second barrel surface **121B**.

(44) The barrel body **121** may include an insulating material. For example, the insulating material may include, but not be limited to, polytetrafluoroethylene (PTFE) (also referred to as Teflon®). Consequently, in an embodiment, the barrel body **121** may exhibit decreased thermal deformation when compared to a related barrel body. Alternatively or additionally, the barrel body **121** may have a decreased heat transfer rate when compared to the related barrel body.

(45) In an embodiment, the floating barrel **120** may include a cavity **122**. The cavity **122** may be configured to accommodate the reaction vessel **102**. In an optional or additional embodiment, the cavity **122** may have a cross-section with a polygonal (e.g., quadrangular) shape. Alternatively or additionally, the cavity **122** may have a substantially circular and/or elliptical cross-section. The cavity **122** may extend from the first barrel surface **121A** to the second barrel surface **121B**. When the reaction vessel **102** is accommodated in the cavity **122**, the reaction vessel **102** may contact the heating plate **112**. The cavity **122** may be disposed on the central portion of the barrel body **121**.

(46) In an embodiment, the floating barrel **120** may include a plurality of irradiators **123**. The plurality of irradiators **123** may be configured to radiate light toward the cavity **122**. The plurality of irradiators **123** may be disposed on respective side barrel surfaces **121C**. For example, a pair of irradiators **123** adjacent to each other may be disposed on side barrel surfaces **121C** (e.g., the side barrel surface **121C** in the +X normal direction and the side barrel surface **121C** in the +Y normal direction) adjacent to each other, respectively. Alternatively or additionally, the plurality of irradiators **123** may be configured to radiate lights in different wavelength bands. For example, one irradiator **123** may be configured to radiate light in one or more wavelength bands and another irradiator **123** may be configured to radiate light in one or more different wavelength bands.

However, the present disclosure is not limited in this regard.

(47) In an embodiment, the irradiator **123** may include a first converging lens **L11** and a first diverging lens **L12** arranged in-line with a light traveling direction (e.g., the -X direction). The first diverging lens **L12** may be closer to the cavity **122** than the first converging lens **L11**. Alternatively, the first converging lens **L11** may be closer to the cavity **122** than the first diverging lens **L12**. Each of the first converging lens **L11** and the first diverging lens **L12** may include, but not be limited to, a bi-convex lens, a plano-convex lens, a meniscus convex lens, other converging lenses, and/or a combination thereof. A combination of the first converging lens **L11** and the first diverging lens **L12** may cause light to be radiated to the reaction vessel **102** in a direction (e.g., the -X direction) substantially orthogonal to an arrangement direction (e.g., the Z direction) of the reaction vessel **102**.

(48) In an embodiment, the irradiator **123** may include a first lens housing **C11**. The first lens housing **C11** may be configured to accommodate the first converging lens **L11**. The first diverging lens **L12** may be disposed inside the barrel body **121** and may be disposed on a first passage **P1** between the side barrel surface **121C** and the cavity **122**. The first diverging lens **L12** may be

disposed outside the first lens housing C11. The first lens housing C11 may have a substantially cylindrical shape. The first lens housing C11 may include cylindrical parts having different diameters and forming a step therebetween. A portion of a cylindrical part having a large diameter of the first lens housing C11 may be disposed on the first passage P1. Alternatively or additionally, another portion of the cylindrical part having a large diameter of the first lens housing C11 may extend from the side barrel surface 121C. In an embodiment, a cylindrical part having a small diameter of the first lens housing C11 may be connected to the cylindrical part having a large diameter and may be disposed outside the barrel body 121.

(49) In an embodiment, the irradiator 123 may include a first connector C12. The first connector C12 may be connected to a light path inside the first lens housing C11. The first connector C12 may be connected to an optical fiber (e.g., the at least one optical fiber F1 of FIG. 1). Light may travel from the light source 11 to the first connector C12 through the optical fiber F1 and may be radiated to the reaction vessel 102 disposed in the cavity 122 through the first converging lens L11 and the first diverging lens L12 along the light path inside the first lens housing C11. A portion of the light radiated to the reaction vessel 102 may be absorbed by a reactant in the reaction vessel 102 and/or another portion of the light may penetrate (e.g., pass-through) the reactant in the reaction vessel 102.

(50) In an embodiment, the irradiator 123 may include a first flange C13. The first flange C13 may be integrally and/or seamlessly connected to the first lens housing C11. The first flange C13 may be detachably coupled to the side barrel surface 121C of the barrel body 121 through one or more fasteners (e.g., a screw). In an optional or additional embodiment, the irradiator 123 may be replaced by another irradiator 123 including a new lens through detachable coupling with the first flange C13.

(51) In an embodiment, the floating barrel 120 may include a plurality of light receivers 124. The plurality of light receivers 124 may be configured to receive light from the cavity 122. The plurality of light receivers 124 may be disposed on corresponding side barrel surfaces 121C. For example, a pair of light receivers 124 adjacent to each other may be disposed on side barrel surfaces 121C (e.g., the side barrel surface 121C in the -X normal direction and the side barrel surface 121C in the -Y normal direction) adjacent to each other, respectively. In an embodiment, each of the plurality of light receivers 124 may be disposed to be opposite to the corresponding irradiator 123. For example, each of the plurality of light receivers 124 may be aligned with the corresponding irradiator 123. The plurality of light receivers 124 may be configured to receive lights in different wavelength bands. For example, one light receiver 124 may be configured to receive light in one or more wavelength bands and another receiver 124 may be configured to receive light in one or more different wavelength bands. However, the present disclosure is not limited in this regard.

(52) In an embodiment, the light receiver 124 may include a second converging lens L21 and a second diverging lens L22 arranged in-line with a direction (e.g., the +X direction) opposite to the light traveling direction. The second diverging lens L22 may be closer to the cavity 122 than the second converging lens L21. Alternatively, the second converging lens L21 may be closer to the cavity 122 than the second diverging lens L22. Each of the second converging lens L21 and the second diverging lens L22 may include, but not be limited to, a bi-convex lens, a plano-convex lens, a meniscus convex lens, other converging lenses, and/or a combination thereof. A combination of the second converging lens L21 and the second diverging lens L22 may cause the reaction vessel 102 to receive light in a direction (e.g., the -X direction) substantially orthogonal to an arrangement direction (e.g., the Z direction) of the reaction vessel 102.

(53) In an embodiment, the light receiver 124 may include a second lens housing C21. The second lens housing C21 may be configured to accommodate the second converging lens L21. The second diverging lens L22 may be disposed inside the barrel body 121. Alternatively or additionally, the second diverging lens L22 may be disposed on a second passage P2 between the side barrel surface

121C and the cavity **122**. In an operational or additional embodiment, the second diverging lens **L22** may be disposed outside the second lens housing **C21**. The second lens housing **C21** may have a substantially cylindrical shape. For example, the second lens housing **C21** may have cylindrical parts having different diameters and forming a step therebetween. A portion of a cylindrical part having a large diameter of the second lens housing **C21** may be disposed on the second passage **P2**. The second passage **P2** may be on substantially the same line as the first passage **P1**. Alternatively or additionally, another portion of the cylindrical part having a large diameter of the second lens housing **C21** may extend from the side barrel surface **121C**. A cylindrical part having a small diameter of the second lens housing **C21** may be connected to the cylindrical part having a large diameter and may be disposed outside the barrel body **121**.

(54) In an embodiment, the light receiver **124** may include a second connector **C22**. The second connector **C22** may be connected to a light path inside the second lens housing **C21**. The second connector **C22** may be connected to an optical fiber (e.g., the at least one optical fiber **F2** of FIG. **1**). The light penetrating (e.g., passing through) a reactant in the reaction vessel **102** may travel to the second connector **C22** through the second diverging lens **L22** and the second converging lens **L21** along the light path provided in the second lens housing **C21** and may travel to the spectroscope **13** through the optical fiber **F2**.

(55) In an embodiment, the light receiver **124** may include a second flange **C23**. The second flange **C23** may be integrally and/or seamlessly connected to the second lens housing **C21**. Alternatively or additionally, the second flange **C23** may be detachably coupled to the side barrel surface **121C** of the barrel body **121** through one or more fasteners (e.g., a screw). In an optional or additional embodiment, the light receiver **124** may be replaced by another light receiver **124** including a new lens through detachable coupling with the second flange **C23**.

(56) In an embodiment, the reactor **101** may include a support body **130**. The support body **130** may be configured to support the floating barrel **120**. The floating barrel **120** may maintain a floating state physically and/or thermally separated from the heating stirrer **110** as the floating barrel **120** is supported by the support body **130**.

(57) In an embodiment, the support body **130** may include a holder **131**. The holder **131** may be configured to hold the barrel body **121**. Alternatively or additionally, the holder **131** may surround the plurality of side barrel surfaces **121C** of the barrel body **121**. The holder **131** may be coupled (e.g., fastened) to the plurality of side barrel surfaces **121C** of the barrel body **121**. The holder **131** may include passage openings through which the plurality of irradiators **123** and the plurality of light receivers **124** disposed in the barrel body **121** pass.

(58) In an embodiment, the support body **130** may include a plurality of guide rails **132**. The plurality of guide rails **132** may be spaced apart from corner regions of the holder **131**. Alternatively or additionally, the plurality of guide rails **132** may be disposed on corresponding corner regions of the holder **131**. The plurality of guide rails **132** may have a shape (e.g., a shaft shape) extending in a longitudinal direction. For example, the plurality of guide rails **132** may extend in the Z-axis direction. Each of the plurality of guide rails **132** may include a screw thread formed on the outer surface of the guide rail **132**.

(59) In an embodiment, the plurality of guide rails **132** may be configured to guide liner movement of the holder **131**. For example, as movement of the holder **131** in the Z-axis direction may be guided by the plurality of guide rails **132**, the size of the gap **G** (e.g., a distance) between the second barrel surface **121B** of the barrel body **121** and a surface (e.g., the surface in the +Z normal direction) of the heating plate **112** may be adjusted.

(60) In an optional or additional embodiment, the plurality of guide rails **132** may guide tilting of the holder **131**. The plurality of guide rails **132** may be connected to corresponding corner regions of the holder **131**. Consequently, the plurality of guide rails **132** may adjust a posture of the holder **131** relative to the Z-axis by independently adjusting the height (e.g., displacement in the Z-axis direction) of the corresponding corner regions of the holder **131**. For example, to tilt the holder **131**

by +45 degrees relative to the X-axis and +45 degrees relative to the Y-axis, when a position of a corner region of the holder **131** is fixed relative to the guide rail **132** that is arranged in a direction of +45 degrees relative to the X-axis and +45 degrees relative to the Y-axis, positions of the corner regions of the holder **131** relative to the other guide rails **132** may be set to be higher than the fixed position.

(61) In an embodiment, the support body **130** may include a plurality of legs **133**. The plurality of legs **133** may be configured to connect the holder **131** to the corresponding guide rail **132**. For example, each of the plurality of legs **133** may extend from the corresponding corner region of the holder **131** to the guide rail **132**.

(62) In an embodiment, the support body **130** may include a plurality of sliders **134**. Each of the plurality of sliders **134** may be configured to slide on the corresponding guide rail **132**. Each of the sliders **134** may be connected to the corresponding leg **133**. The holder **131**, the plurality of legs **133**, and the plurality of sliders **134** may be integrally and/or seamlessly formed.

(63) In an embodiment, each of the plurality of sliders **134** may include a tab hole H through which the corresponding guide rail **132** passes. The tab hole H may include a screw thread formed on the inner surface of the tab hole. For example, the screw thread may be configured to engage with a screw thread formed on the outer surface of the guide rail **132**. Fine adjustment of displacement of the slider **134** may be implemented by engagement of the screw thread of the tab hole H with the screw thread of the guide rail **132**.

(64) In an embodiment, the support body **130** may include a base **135**. The base **135** may function as a pedestal configured to support the driving unit **111** and the plurality of guide rails **132**. For example, the driving unit **111** may be disposed on a central region of the base **135** and the plurality of guide rails **132** may be disposed on corresponding corner regions of the base **135**. Alternatively or additionally, the plurality of guide rails **132** may be separated from the driving unit **111** on the base **135**.

(65) In an embodiment, the support body **130** may include a knob **136**. The knob **136** may be configured to adjust a position of the slider **134** on the guide rail **132**. Alternatively or additionally, the knob **136** may be coupled to an end of the guide rail **132**. For example, the knob **136** may adjust a position of the slider **134** on the guide rail **132** by rotating the guide rail **132**. Position adjustment of the slider **134** may lead to adjustment of a position of the leg **133** corresponding to the slider **134**, and position adjustment of the leg **133** may lead to position adjustment of a corner region of the holder **131** connected to the leg **133**.

(66) In an embodiment, the support body **130** may include one guide rail, one leg, and/or one slider.

(67) In an embodiment, the optical monitoring device **100** may include a chiller **103**. The chiller **103** may be configured to chill the reaction vessel **102**.

(68) In an embodiment, the chiller **103** may include a chilling block **1031**. The chilling block **1031** may be disposed on and/or above the floating barrel **120** and the holder **131**.

(69) In an embodiment, the chiller **103** may include an accommodating opening **1032**. The accommodating opening **1032** may be configured to accommodate the reaction vessel **102**. The accommodating opening **1032** may be disposed in a central region of the chilling block **1031**. Alternatively or additionally, the accommodating opening **1032** may be substantially aligned with the cavity **122**.

(70) In an embodiment, the chiller **103** may include a plurality of coupling openings **1033**. Each of the plurality of coupling openings **1033** may be configured to be coupled to a corresponding guide rail **132**. Alternatively or additionally, each of the plurality of coupling openings **1033** may be configured to be coupled to the knob **136**. The plurality of coupling openings **1033** may be disposed in corresponding corner regions of the chilling block **1031**.

(71) In an embodiment, the chiller **103** may include a plurality of chilling passages **1034**. The plurality of chilling passages **1034** may be configured to accommodate a cooling liquid conduit from a circulator (e.g., circulator **15** of FIG. **1**) and a cooling liquid conduit leading to the circulator

15. Heat exchange with the reaction vessel **102** may be performed through the chilling block **1031** from cooling liquid conduits respectively accommodated in the plurality of chilling passages **1034**. The temperature of the reaction vessel **102** may be maintained to be in a determined temperature range. A vaporized substance in the reaction vessel **102** may condense. The condensation may prevent a decrease in a level of a solution in the reaction vessel **102** as a solvent in the reaction vessel **102** boils and vaporizes when monitoring a long-term reaction.

(72) FIG. **8** is a perspective view of a jig, according to an embodiment.

(73) Referring to FIG. **8**, the optical monitoring device **100** may include the heating stirrer **110**. The heating stirrer **110** may include a heating plate **112** and a jig **113**. The jig **113** may be configured to fix the reaction vessel **102** in place. In an embodiment, the jig **113** may be configured to separate the reaction vessel **102** from the barrel body **121** of FIGS. **2** to **7**. The reaction vessel **102**, which may be separated from the barrel body **121**, may not contact the barrel body **121** by the jig **113**. Alternatively or additionally, the jig **113** may be disposed inside the cavity **122** of FIGS. **2** to **7**.

(74) In an embodiment, the jig **113** may include a jig base **B1**. The jig base **B1** may protrude in a direction (e.g., the +Z direction) from a region (e.g., a central region) of a surface (e.g., the surface in the +Z normal direction) of the heating plate **112**. In an optional or additional embodiment, the jig base **B1** may have a substantially cylindrical shape.

(75) In an embodiment, the jig **113** may include a jig body **B2**. The jig body **B2** may include a jig end surface **B21** opposite to a surface (e.g., the surface in the +Z normal direction) of the jig base **B1** and a plurality of jig side surfaces **B22** between the jig end surface **B21** and the surface of the jig base **B1**. In an optional or additional embodiment, the jig body **B2** may have a substantially polygonal shape. The plurality of jig side surfaces **B22** and a side surface of the jig base **B1** may form a gap.

(76) In an embodiment, the jig **113** may include a hollow portion **B3**. The hollow portion **B3** may be configured to accommodate the reaction vessel **102**. The hollow portion **B3** may extend from the jig end surface **B21** to another surface (e.g., the surface in the -Z normal direction) of the jig base **B1**. Alternatively or additionally, the hollow portion **B3** may be formed across the jig base **B1** and the jig body **B2**. The reaction vessel **102** disposed in the hollow portion **B3** may contact the heating plate **112**.

(77) In an embodiment, the jig **113** may include a plurality of first slits **SL1**. Each of the plurality of first slits **SL1** may be configured to align with a corresponding irradiator **123** (e.g., the first diverging lens **L12** of FIGS. **2** to **7**). For example, the light radiated from the irradiator **123** may pass the first slit **SL1** and travel to the reaction vessel **102**. The spacing between the first slits **SL1** may be selected to pass light in a determined wavelength band. In an optional or additional embodiment, the plurality of first slits **SL1** may be disposed on corresponding jig side surfaces **B22**. For example, one of the first slits **SL1** may be disposed on a jig side surface **B22** in the +X direction and another first slit **SL1** may be disposed on a jig side surface **B22** in the -X direction.

(78) In an optional or additional embodiment, the jig **113** may include a plurality of second slits **SL2**. Each of the second slits **SL2** may be configured to align with a corresponding light receiver **124** (e.g., the second diverging lens **L22** of FIGS. **2** to **7**). For example, the light penetrating (e.g., passing through) a substance in the reaction vessel **102** may pass the second slit **SL2** and may travel to the light receiver **124**. The spacing between the second slits **SL2** may be selected to pass light in a determined wavelength band. In an optional or additional embodiment, each of the plurality of second slits **SL2** may be disposed on a jig side surface **B22** opposite to the jig side surface **B22** on which the corresponding first slit **SL1** is disposed. For example, one of the second slits **SL2** may be disposed on a jig side surface **B22** in the -X direction opposite to the jig side surface **B22** in the +X direction on which the corresponding first slit **SL1** is disposed and another second slit **SL2** may be disposed on a jig side surface **B22** in the -Y direction opposite to the jig side surface **B22** in the +Y direction on which the corresponding first slit **SL1** is disposed.

(79) In an embodiment, the jig **113** may include a thermally conductive material. For example, the

jig **113** may include a same material (e.g., a metal material) as the heating plate **112**. Alternatively or additionally, the jig **113** may include a different material as the heating plate **112**. In an embodiment, heat transferred to the jig **113** from the heating plate **112** may be conducted to the reaction vessel **102**. The jig **113** may be integrally and seamlessly connected to the heating plate **112**.

(80) FIG. **8** illustrates the irradiator **123** arranged in the +X direction and the light receiver **124** arranged in the -Y direction. However, this is an example for describing the shape of the jig **113** and is not limited thereto. For example, a light receiver arranged in the -X direction opposite to the irradiator **123** and an irradiator arranged in the +Y direction opposite to the light receiver **124** may be disposed on corresponding slits, without departing from the scope of the present disclosure.

(81) FIG. **9** is a diagram illustrating a floating barrel at a predetermined distance from a heating plate, according to an embodiment.

(82) Referring to FIG. **9**, a floating barrel A2 may be disposed with a gap G of approximately 5 millimeters (mm) from a heating plate A1. Heat transfer between the heating plate A1 and the floating barrel A2 may include heat transfer by convection. In an example depicted by FIG. **9**, a coefficient of thermal expansion (CTE) of the floating barrel A2 may be selected to be approximately $135 \times 10^{-6} \text{ mm}/^\circ \text{C}$. In such an example, a height h1 of the floating barrel A2 may be determined to be approximately 10 mm. Alternatively or additionally, a height h2 of a lens may be determined to be approximately 0.4 mm. That is, heights h2 of a first lens L1 (e.g., an irradiator lens) and a second lens L2 (e.g., a light receiver lens) arranged in the floating barrel A2 may be determined to be approximately 0.4 mm. A temperature change $\Delta T1$ at a first point T1 where the first lens L1 is disposed may be approximately 75 degrees Celsius ($^\circ \text{C}$.) and a temperature change $\Delta T2$ at a second point T2 where the second lens L2 is disposed and apart from the first lens L1 by a distance d may be approximately 76°C .

(83) FIG. **10** is a diagram illustrating thermal deformation of a floating barrel at a predetermined distance from a heating plate, according to an embodiment. FIG. **11** is a diagram illustrating a height deviation of lenses, according to an embodiment.

(84) Referring to FIGS. **10** and **11**, a thermal deformation amount δ at a point of the floating barrel A2 on which lenses L1 and L2 are disposed may be determined using an equation like $\delta = \alpha \times \Delta T \times h1$, where α represents the CTE of the floating barrel A2, ΔT represents the temperature change at the point of the floating barrel, and h1 represents the height of the floating barrel A2. For example, a thermal deformation amount $\delta1$ of a first point T1 on the floating barrel A2 on which the irradiator lens L1 is disposed may be approximately 0.10125 mm and a thermal deformation amount $\delta2$ of a second point T2 on the floating barrel A2 on which the light receiver lens L2 is disposed may be about 0.10260 mm. Thus, referring to FIG. **11**, a height deviation **48** due to thermal deformation of the floating barrel A2 may be approximately 0.00135 mm, and a deviation angle θ may be approximately 0.00155 degrees. Because heights (h2 of FIG. **9**) of the lenses L1 and L2 are approximately 0.4 mm, a height deviation of the lenses L1 and L2 due to thermal deformation may be calculated to be approximately 1.35 micrometers (μm). Consequently, a light loss rate due to thermal deformation may be approximately 0.338%.

(85) FIG. **12** are graphs showing a height deviation due to thermal deformation (**48**) and a light loss rate due to the thermal deformation based on a distance between a heating plate and a floating barrel (e.g. gap G of FIG. **6**), according to an embodiment.

(86) Referring to FIG. **12** and Table 1, the height deviation and the light loss rate may be determined at the following distances: (i) at 0 mm (when a floating barrel contacts a heating plate), (ii) at approximately 1 mm, (iii) at approximately 3 mm, (iv) at approximately 5 mm, and (v) at approximately 10 mm. T1 may represent a temperature at a first point in which a first lens (e.g., the first lens L1 of FIGS. **9** to **11**) is disposed and T2 may represent a temperature at a second point in which a second lens (e.g., the second lens L2 of FIGS. **9** to **11**) is disposed. A PTFE having a CTE of approximately $135 \times 10^{-6} \text{ mm}/^\circ \text{C}$ may be selected as a material of the floating barrel.

(87) TABLE-US-00001 TABLE 1 Height Deviations and Light Loss Rates for PTFE. Height Light Distance T1 T2 Deviation Loss Rate (mm) (° C.) (° C.) ($\Delta\delta$) (μm) (%) 0 200.0 250.0 67.50 16.8750 1 180.0 220.2 54.27 13.5675 3 140.0 160.6 27.81 6.9500 5 100.0 101.0 1.35 0.3375 10 30.0 30.2 0.27 0.0675

(88) As shown in FIG. 12 and Table 1, as the distance increases, the height deviation of lenses due to thermal deformation may decrease, and thus, the light loss rate may decrease. For example, when the distance between the floating barrel and the heating plate is greater than or equal to approximately 5 mm, the light loss rate may be less than 1% (e.g., approximately a 99% confidence level). To achieve a light loss rate at a 95% confidence level, the distance may need to be greater than or equal to approximately 3 mm. Alternatively or additionally, when the floating barrel does not float from the heating plate and contacts the heating plate (e.g., distance is approximately 0 mm), the light loss rate may significantly increase and the measurement accuracy and repeatability of the optical monitoring device may decrease.

(89) In an embodiment, an upper limit of the distance between the floating barrel 120 and the heating plate 112 may be determined based on an arrangement environment of the optical monitoring device (e.g., optical monitoring device 10 of FIG. 1, optical monitoring device 100 of FIGS. 2 to 8). For example, the upper limit of the distance between the floating barrel 120 and the heating plate 112 may be approximately 10 cm or less, approximately 8 cm or less, approximately 5 cm or less, approximately 3 cm or less, or approximately 2 cm or less.

(90) FIG. 13 is a perspective view of a floating barrel, according to an embodiment. An optical monitoring device 100-1 of FIG. 13 may include or may be similar in many respects to at least one of the optical monitoring device 10 and the optical monitoring device 100 described above with reference to FIGS. 1 to 8, and may include additional features not mentioned above.

(91) Referring to FIG. 13, the optical monitoring device 100-1 may include a floating barrel 120-1. The floating barrel 120-1 may include a barrel body 121 and a cavity 122. The barrel body 121 may include a first barrel surface 121A, a second barrel surface 121B, and a plurality of side barrel surfaces 121C.

(92) In an embodiment, the floating barrel 120-1 may include an access area 125. The access area 125 may be configured to allow access of a reaction vessel 102 from the outside of the barrel body 121 to inside the cavity 122. Alternatively or additionally, the access area 125 may be configured to allow access of the reaction vessel 102 from the cavity 122 to the outside of the barrel body 121. The access area 125 may be formed by opening at least a portion of regions of one of the plurality of side barrel surfaces 121C. A numerical value (e.g., a length, a width) of the access area 125 may be substantially the same as a numerical value (e.g., a length, a width) of the cavity 122. That is, dimensions of the access area 125 may be similar or substantially the same as corresponding dimensions of the cavity 122. The reaction vessel 102 may be separated from the barrel body 121 through the access area 125 from the cavity 122. In an optional or additional embodiment, a new reaction vessel may be disposed inside the cavity 122 through the access area 125.

(93) In an embodiment, the optical monitoring device 100-1 may include a support body 130-1. The support body 130-1 may include a shaft 137 and an arm 138. The shaft 137 may include a fixed axis (e.g., the Z-axis) extending in a longitudinal direction. The arm 138 may be configured to connect the shaft 137 to the barrel body 121. For example, a first end of the arm 138 may be connected to the shaft 137 and a second end of the arm 138 that is opposite to the first end may be connected to a region (e.g., a corner region) of one of the side barrel surfaces 121C. Alternatively or additionally, the arm 138 may be configured to rotate on the shaft 137. For example, the barrel body 121 may have a configuration disposed in one position on a heating stirrer (e.g., the heating stirrer 110 of FIGS. 2 to 7) by rotation of the arm 138 on the shaft 137. Alternatively or additionally, the barrel body 121 may have a configuration disposed on a position offset from the aforementioned position on the heating stirrer 110.

(94) FIG. 14 is a perspective view of a linear guide, according to an embodiment. FIG. 15 is a

perspective view of a structure for fixing and releasing a lens, according to an embodiment. An optical monitoring device **100-2** of FIG. **14** may include or may be similar in many respects to at least one of the optical monitoring device **10**, the optical monitoring device **100**, and the optical monitoring device **100-1** described above with reference to FIGS. **1** to **8** and **13**, and may include additional features not mentioned above.

(95) Referring to FIGS. **14** and **15**, the optical monitoring device **100-2** may include a floating barrel **120-2**. The floating barrel **120-2** may include a barrel body **121-2** and an irradiator **123-2**. The irradiator **123-2** may include a lens housing **C1** (e.g., a first lens housing **C11** of FIGS. **2** to **7**) and at least one lens **L** (e.g., a converging lens and/or a diverging lens).

(96) In an embodiment, the barrel body **121-2** may include a plurality of first linear guides **G1**. The plurality of first linear guides **G1** may be disposed on the inner surface of a passage facing the irradiator **123-2**. Alternatively or additionally, the plurality of first linear guides **G1** may extend in a direction (e.g., a direction toward the cavity **122** of FIGS. **2** to **7**, a direction away from the cavity **122**, and/or $+X$ directions) along the inner surface of the passage. Each of the plurality of first linear guides **G1** may include a rib shape. In an embodiment, the plurality of first linear guides **G1** may be spaced apart from one another in another direction (e.g., the $\pm Z$ directions). In an optional or additional embodiment, the plurality of first linear guides **G1** may include first linear guides **G1** of a first group facing a first side of the irradiator **123-2** and first linear guides **G1** of a second group facing a second side opposite to the first side of the irradiator **123-2**.

(97) In an optional or additional embodiment, the irradiator **123-2** may include a plurality of second linear guides **G2**. The plurality of second linear guides **G2** may be configured to engage with the plurality of first linear guides **G1**. Consequently, the mutual engagement of the plurality of first linear guides **G1** and the plurality of second linear guides **G2** may reduce and/or prevent an offset of an optical axis of a lens **L**.

(98) In an embodiment, the movement of the plurality of second linear guides **G2** may be guided along the plurality of first linear guides **G1**. As the second linear guides **G2** are guided by the first linear guides **G1**, the lens **L** may be configured to move in a direction (e.g., the $-X$ direction) toward the cavity **122** of FIGS. **2** to **7** and/or a direction (e.g., the $+X$ direction) away from the cavity **122**. Alternatively or additionally, the movement of the plurality of second linear guides **G2** may be guided in an extending direction (e.g., the $+X$ directions) of the plurality of first linear guides **G1**.

(99) In an embodiment, each of the plurality of second linear guides **G2** may include a rib shape. The plurality of second linear guides **G2** may be disposed on the plurality of first linear guides **G1**. Alternatively or additionally, the plurality of second linear guides **G2** may be disposed on an outer surface of the lens housing **C1** of the irradiator **123-2**. In an embodiment, the plurality of second linear guides **G2** may be spaced apart in a distancing direction (e.g., $+Z$ directions) of the plurality of first linear guides **G1**. In some embodiments, the number of second linear guides **G2** may be greater than the number of first linear guides **G1**. In an optional or additional embodiment, the plurality of second linear guides **G2** may include a first group of second linear guides **G2** configured to be guided by the first group of the first linear guides **G1** and a second group of second linear guides **G2** configured to be guided by the second group of the first linear guides **G1**.

(100) In an embodiment, the floating barrel **120-2** may include a plurality of actuators **126**. The actuator **126** may be configured to secure the lens housing **C1** to the barrel body **121-2**. Alternatively or additionally, the actuator **126** may be configured to release the lens housing **C1** from the barrel body **121-2**. The lens **L** in the lens housing **C1** may be replaced as the lens housing **C1** is separated from the barrel body **121** by the actuator **126**, a new lens may be disposed in the lens housing **C1**, and then the lens housing **C1** may be secured to the barrel body **121** by the actuator **126**.

(101) Referring to FIG. **15**, the actuator **126** may include a slider **D1**. The slider **D1** may be configured to slide in one or more directions (e.g., the $+X$ directions, $+Y$ directions) relative to the

barrel body **121-2**. The slider **D1** may include a base arm **D11**. In some embodiments, the slider **D1** may include a first extending arm **D12** extending in a direction (e.g., the +Y direction) from a first end (e.g., an end in the -Z direction) of the base arm **D11**. Alternatively or additionally, the slider **D1** may include a second extending arm **D13** extending in another direction (e.g., the -X direction) from an end opposite to the end connected to the first end of the base arm **D11** among ends of the first extending arm **D12**. In an optional or additional embodiment, the slider **D1** may include a third extending arm **D14** extending in a direction (e.g., the +Y direction) from a second end (e.g., an end in the +Z direction) of the base arm **D11**. In another optional or additional embodiment, the slider **D1** may include a protrusion **D15** protruding from a surface (e.g., the surface in the -Y normal direction) of the base arm **D11**.

(102) In an embodiment, the actuator **126** may include a tab **D2**. The tab **D2** may be configured to contact the lens housing **C1** and press the lens housing **C1**. When the lens housing **C1** is disposed in a passage in the lens barrel **121-2**, the lens housing **C1** may be pressed by the tab **D2**. A surface (e.g., the surface substantially in the +Y normal direction) of the tab **D2** contacting the lens housing **C1** may have a curved surface. Alternatively or additionally, the tab **D2** may be disposed in a second slot **S2** formed in the inner surface (e.g., the surface in the +Y normal direction) of the barrel body **121-2**. In an optional or additional embodiment, the movement of the tab **D2** may be guided in a longitudinal direction (e.g., the +X directions) of the second slot **S2**. In another optional or additional embodiment, the tab **D2** may be connected to an end of the second extending arm **D13**.

(103) In an optional or additional embodiment, the actuator **126** may include a handle **D3**. The handle **D3** may be configured to transmit an external force (e.g., a gripping force of a user) to the slider **D1**. For example, when a force is applied to the handle **D3**, the lens housing **C1** may be separated from the barrel body **121-2**. In some embodiments, the handle **D3** may have a button shape. The handle **D3** may be disposed in a first slot **S1** formed in a surface (e.g., the surface in the +Z normal direction) of the barrel body **121-2**. In an optional or additional embodiment, the movement of the handle **D3** may be guided in a longitudinal direction (e.g., the +Y direction) of the first slot **S1**. The handle **D3** may be connected to an end region of the third extending arm **D14**. In another optional or additional embodiment, the handle **D3** may protrude in a direction (e.g., the +Z direction) from the third extending arm **D14**.

(104) In an embodiment, the actuator **126** may include an elastic body **D4**. The elastic body **D4** may elastically support the slider **D1**. Elastic support of the slider **D1** may provide elasticity to the tab **D2**. For example, when the handle **D3** is pulled in the -Y direction, the elastic body **D4** may be compressed by the slider **D1**. Alternatively or additionally, when the handle **D3** is pushed in the +Y direction, the elastic body **D4** may be decompressed. In some embodiments, the elastic body **D4** may include a compression spring. The elastic body **D4** may be disposed to enclose the protrusion **D15**.

(105) Referring to FIG. **14**, one of the actuators **126** may be disposed on a portion of the barrel body **121-2** facing the first side of the lens housing **C1** and another actuator **126** may be disposed on a portion of the barrel body **121-2** facing the second side opposite to the first side of the lens housing **C1**.

(106) In an embodiment, the floating barrel **120-2** may include a light receiver having a similar structure to the irradiator **123-2**. The light receiver may include at least one linear guide that is substantially similar and/or the same as the second linear guide **G2**.

(107) FIG. **16** is a perspective view of a structure for fixing and releasing a lens, according to an embodiment. An actuator **126-1** of FIG. **16** may include or may be similar in many respects to the actuator **126** described above with reference to FIGS. **14** and **15**, and may include additional features not mentioned above.

(108) Referring to FIG. **16**, the actuator **126-1** may include a slider **D1**, a tab **D2**, a handle **D3**, a protrusion **D15**, and an elastic body **D4**. The slider **D1** may include a base arm **D11**. The tab **D2**

may be disposed in a first side region (e.g., a region in the +Y normal direction adjacent to the surface in the -Z normal direction) adjacent to the first end region of the base arm D11. The handle D3 may be disposed in a second end region (e.g., a surface in the +Z normal direction) of the base arm D12. The protrusion D15 may be disposed on a second side region (e.g., a region in the -Y normal direction) opposite to the first side region of the base arm D11.

(109) FIG. 17 is a diagram schematically illustrating an optical monitoring device, according to an embodiment. An optical monitoring device 100-3 of FIG. 17 may include or may be similar in many respects to at least one of the optical monitoring device 10, the optical monitoring device 100, the optical monitoring device 100-1, and the optical monitoring device 100-2 described above with reference to FIGS. 1 to 8 and 13 to 14, and may include additional features not mentioned above.

(110) Referring to FIG. 17, the optical monitoring device 100-3 may be configured to monitor a chemical change of a reactant in a conduit 102-3 in real-time. The reactant may flow along the conduit 102-3.

(111) In an embodiment, the optical monitoring device 100-3 may include a heating stirrer 110-3. The heating stirrer 110-3 may have a shape enclosing the conduit 102-3. The heating stirrer 110-3 may be configured to transfer thermal energy and/or stirring energy to both sides of the conduit 102-3. For example, the heating stirrer 110-3 may include a first heating stirrer 110-31 disposed on a first side (e.g., the upper side of FIG. 17) of the conduit 102-3 and a second heating stirrer 110-32 disposed on a second side (e.g., the lower side of FIG. 17) opposite to the first side of the conduit 102-3. Alternatively or additionally, the first heating stirrer 110-31 and the second heating stirrer 110-32 may extend along the length of the conduit 102-3.

(112) In an embodiment, the optical monitoring device 100-3 may include an irradiator 123-3 and a light receiver 124-3. The irradiator 123-3 may be disposed on the first side (e.g., the first heating stirrer 110-31) of the heating stirrer 110-3. The light receiver 124-3 may be disposed on the second side (e.g., the second heating stirrer 110-32) of the heating stirrer 110-3. In some embodiments, the irradiator 123-3 and the light receiver 124-3 may be thermally and/or physically separated from the heating stirrer 110-3 in a similar manner described with reference to FIGS. 2 to 7. Alternatively or additionally, the irradiator 123-3 and the light receiver 124-3 may form a light path at a predetermined position of the conduit 102-3.

(113) FIG. 18 is a graph showing a light loss factor based on a distance between a heating plate and a floating barrel, according to an embodiment.

(114) Referring to FIG. 18 and Table 2, the light loss rate may be determined at the following distances: (i) at 0 mm (when a floating barrel contacts a heating plate), (ii) at approximately 1 mm, (iii) at approximately 2 mm, (iv) at approximately 3 mm, (v) at approximately 5 mm, and (vi) at approximately 10 mm. T1 may represent a temperature at a first point in which a first lens (e.g., the first lens L1 of FIGS. 9 to 11) is disposed and T2 may represent a temperature at a second point in which a second lens (e.g., the second lens L2 of FIGS. 9 to 11) is disposed. Materials of the floating barrels may be selected to be PTFE and glass wool having CTEs of approximately $135 \times 10^{-6} \text{ mm}/^\circ \text{C}$. and $9.694 \times 10^{-6} \text{ mm}/^\circ \text{C}$., respectively. That is, the thermal conductivity of glass wool may be less than the thermal conductivity of PTFE.

(115) TABLE-US-00002 TABLE 2 Height Deviations and Light Loss Rates for Glass Wool. Height Light Distance T1 T2 Deviation Loss Rate (mm) ($^\circ \text{C}$.) ($^\circ \text{C}$.) ($\Delta\delta$) (μm) (%) 0 50.0 90.0 3.88 0.9694 1 47.0 77.2 2.93 0.7319 2 43.0 64.0 2.93 0.5089 3 40.0 53.2 1.47 0.3199 5 34.0 34.5 0.09 0.0242 10 30.0 30.2 0.02 0.0049

(116) As shown in Table 1, a light loss rate of 0.3375% at a distance of approximately 5 mm may be calculated for PTFE. Alternatively, as shown in FIG. 18 and Table 2, glass wool may have a light loss rate of 0.3199% at a distance of approximately 3 mm. That is, the light loss rate when the material of the floating barrel is glass wool and the distance is set to approximately 3 mm may be similar to the light loss rate when the material of the floating barrel is PTFE and the distance is set

to approximately 5 mm.

(117) While the present disclosure includes specific examples, it is to be apparent to one of ordinary skill in the art that various changes in form and details may be made in these examples without departing from the spirit and scope of the claims and their equivalents. The examples described herein are to be considered in a descriptive sense only, and not for purposes of limitation. Descriptions of features or aspects in each example are to be considered as being applicable to similar features or aspects in other examples. For example, suitable results may be achieved when the described techniques are performed in a different order and/or when components in a described system, architecture, device, or circuit are combined in a different manner and/or replaced or supplemented by other components or their equivalents.

Claims

1. An optical monitoring device, comprising: a heating stirrer; a floating barrel; and a support body arranged to support the floating barrel, wherein the floating barrel comprises: a barrel body facing the heating stirrer and spaced apart from the heating stirrer by a gap; a cavity provided in the barrel body and configured to accommodate at least one of a vessel and a conduit; an irradiator arranged in the barrel body and configured to radiate light to the cavity; and a light receiver arranged in the barrel body, aligned with the irradiator and the cavity, and configured to receive light from the cavity.
2. The optical monitoring device of claim 1, wherein a size of the gap between the barrel body and the heating stirrer is greater than or equal to 3 mm.
3. The optical monitoring device of claim 1, wherein the support body comprises: a holder configured to hold the floating barrel; a plurality of guide rails configured to guide movement of the holder; and a plurality of legs coupling the holder to the plurality of guide rails.
4. The optical monitoring device of claim 3, wherein the support body further comprises a knob configured to adjust a position of a leg of the plurality of legs on a corresponding guide rail of the plurality of guide rails.
5. The optical monitoring device of claim 3, wherein each of the plurality of legs comprises a tab hole configured to pass a corresponding guide rail of the plurality of guide rails.
6. The optical monitoring device of claim 1, wherein the heating stirrer comprises: a heating plate facing the barrel body; and a jig protruding from the heating plate, and configured to separate at least one of the vessel and the conduit, from the barrel body.
7. The optical monitoring device of claim 6, wherein the jig comprises: a first slit aligned with the irradiator; and a second slit aligned with the light receiver.
8. The optical monitoring device of claim 1, wherein the barrel body comprises an insulating material.
9. The optical monitoring device of claim 1, wherein the barrel body comprises: a first passage extending from the cavity and configured to accommodate the irradiator; and a second passage extending from the cavity and configured to accommodate the light receiver.
10. The optical monitoring device of claim 1, wherein each of the irradiator and the light receiver comprises: at least one lens; and a lens housing configured to accommodate the at least one lens; wherein the lens housing is further configured to be detachably coupled to the barrel body.
11. The optical monitoring device of claim 10, wherein the at least one lens comprises a converging lens and a diverging lens.
12. The optical monitoring device of claim 1, further comprising: a light source configured to generate light in a determined wavelength band; and a spectroscope configured to analyze at least a portion of the light generated by the light source and received by the light receiver that has been radiated by the irradiator and has passed through at least one of the vessel and the conduit.
13. The optical monitoring device of claim 1, further comprising: a chiller configured to cool at

least one of the vessel and the conduit.

14. The optical monitoring device of claim 13, further comprising: a circulator configured to circulate a cooling liquid to the chiller.

15. The optical monitoring device of claim 1, wherein the barrel body further comprises an access area open to the cavity.

16. The optical monitoring device of claim 1, further comprising: a shaft; and an arm configured to connect the shaft to the floating barrel and rotate relative to the shaft.

17. The optical monitoring device of claim 1, further comprising: a first linear guide configured to guide linear movement of a first lens of the irradiator in a first direction toward the cavity and a second direction away from the cavity; and a second linear guide configured to guide linear movement of a second lens the light receiver in the first direction toward the cavity and the second direction away from the cavity.

18. The optical monitoring device of claim 1, further comprising: a tab configured to secure a lens of at least one of the irradiator and the light receiver; and an elastic body configured to elastically support the tab.

19. The optical monitoring device of claim 18, further comprising: a slider configured to: slide in a first direction away from the lens; slide in a second direction toward the lens; and deform the elastic body.

20. The optical monitoring device of claim 19, further comprising: a handle coupled to the slider.
